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Strong et al.

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(54) **DIELECTRIC FILM COATING FOR
THROUGH GLASS VIAS AND PLANE
SURFACE ROUGHNESS MITIGATION**

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H01L 23/15 (2006.01)
H01L 23/498 (2006.01)
H01G 4/012 (2006.01)

H01G 4/224 (2006.01)

H01G 4/33 (2006.01)

(52) **U.S. CL.**

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(2013.01); **H01L 23/49816** (2013.01); **H01L**
23/49827 (2013.01); **H01G 4/012** (2013.01);
H01G 4/224 (2013.01); **H01G 4/33** (2013.01)

(58) **Field of Classification Search**

CPC .. **H01L 23/15**; **H01L 23/481**; **H01L 23/49827**
See application file for complete search history.

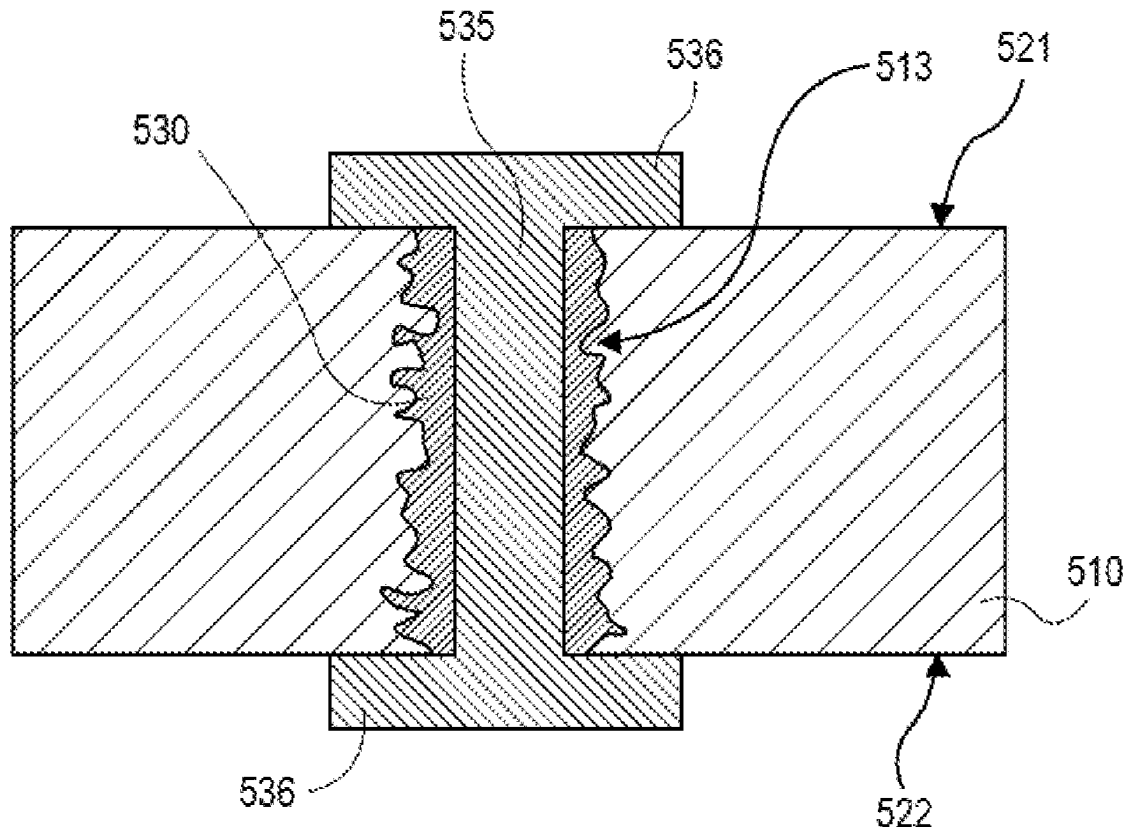
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(57) **ABSTRACT**

Embodiments disclosed herein include electronic packages. In an embodiment, an electronic package comprises a substrate with a first surface and a second surface, where the substrate comprises glass. In an embodiment, the electronic package further comprises a via opening through the substrate, where sidewalls of the via opening have a root mean squared (RMS) surface roughness that is approximately 100 nm or greater. In an embodiment, the electronic package further comprises a liner over the sidewalls of the via opening, where an RMS surface roughness of the liner is approximately 50 nm or smaller. An electronic package may further comprise a via through the via opening.

17 Claims, 16 Drawing Sheets



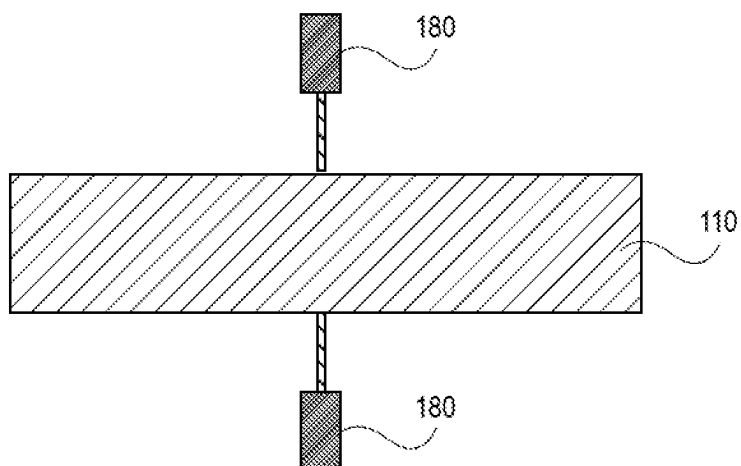


FIG. 1A

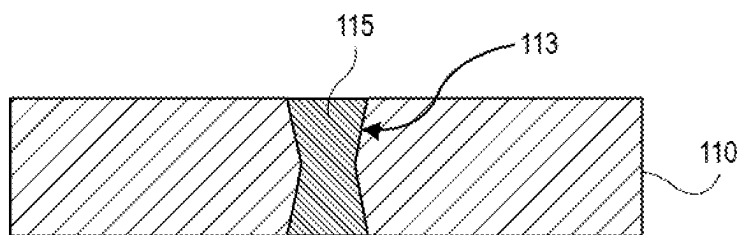


FIG. 1B

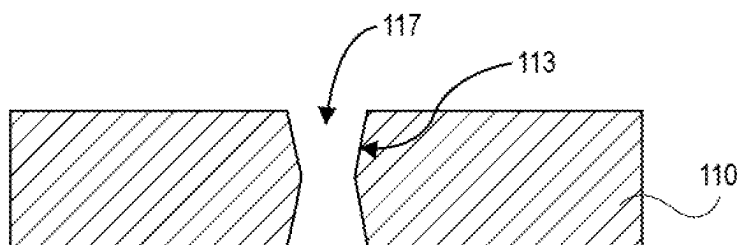


FIG. 1C

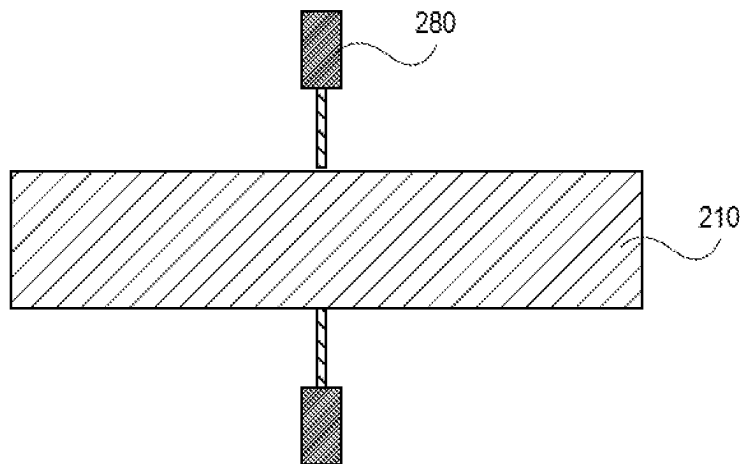


FIG. 2A

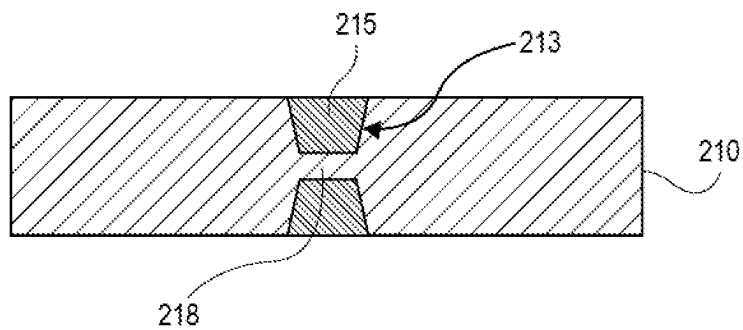


FIG. 2B

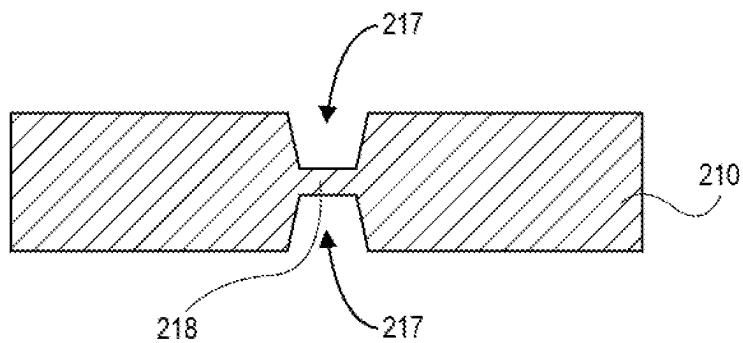


FIG. 2C

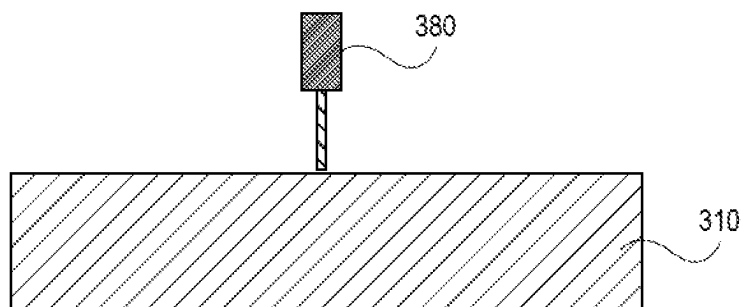


FIG. 3A

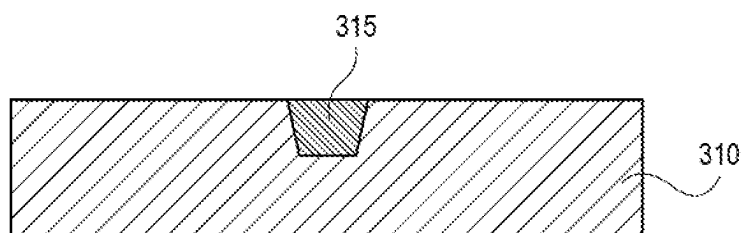


FIG. 3B

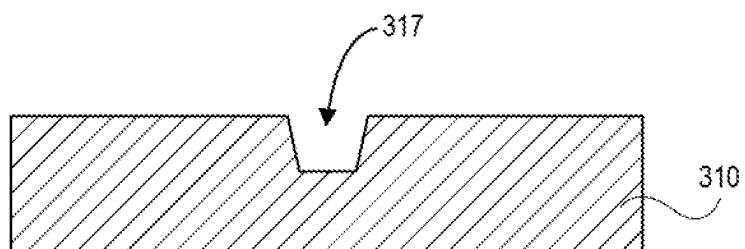


FIG. 3C

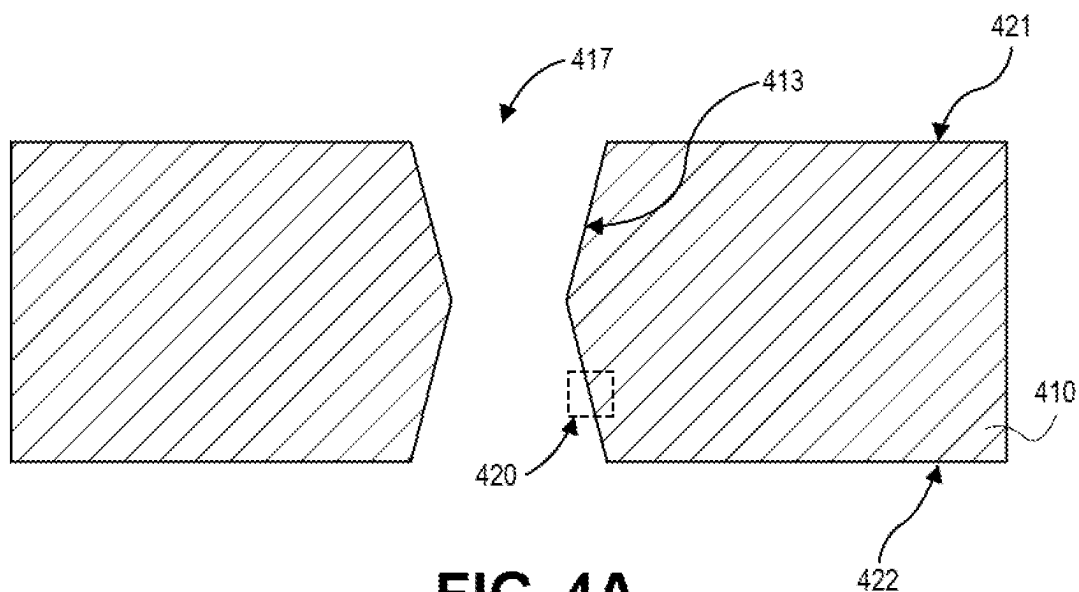


FIG. 4A

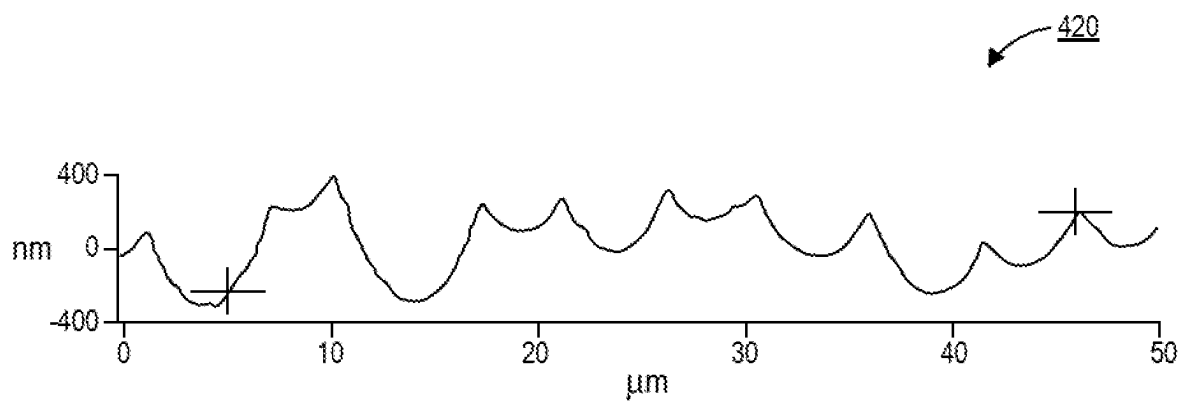


FIG. 4B

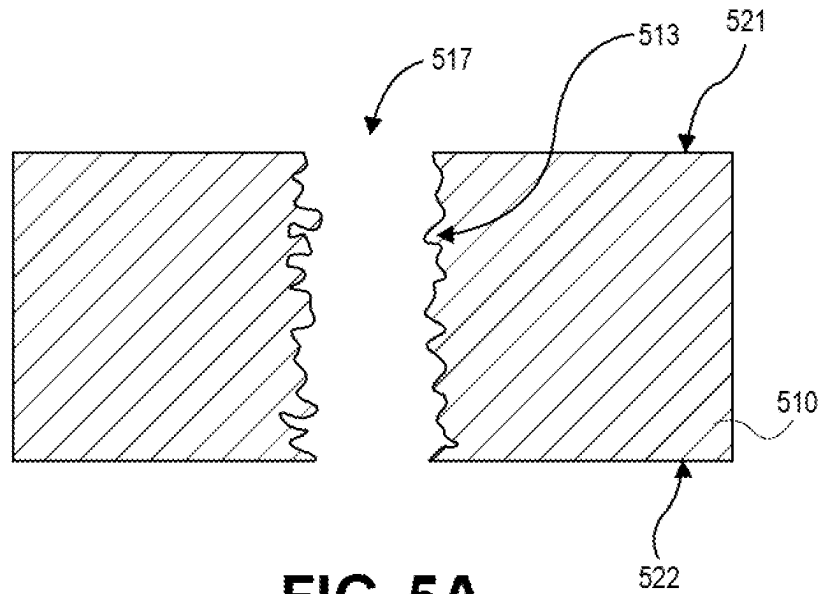


FIG. 5A

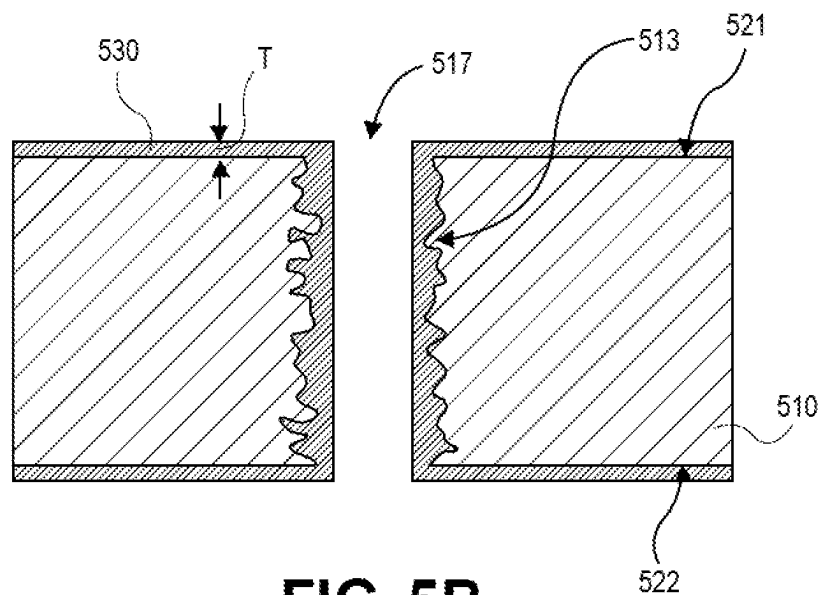
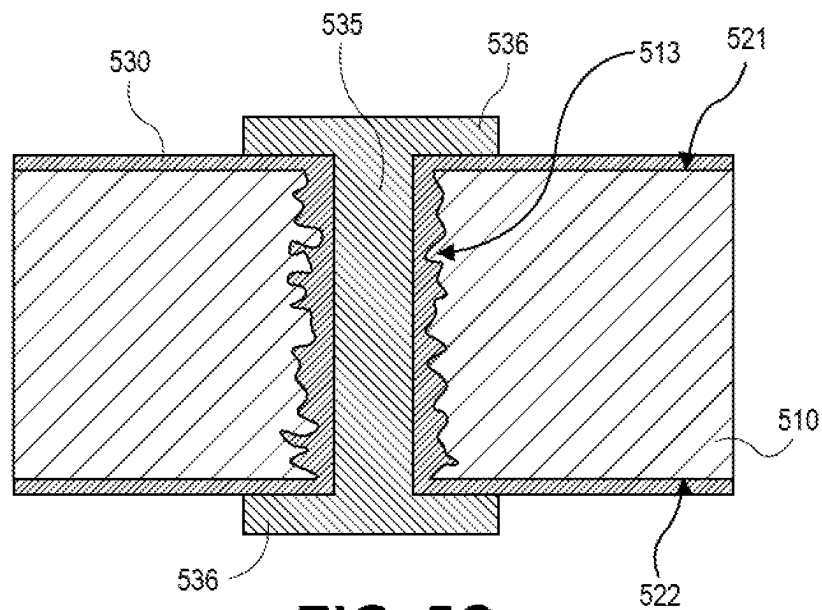
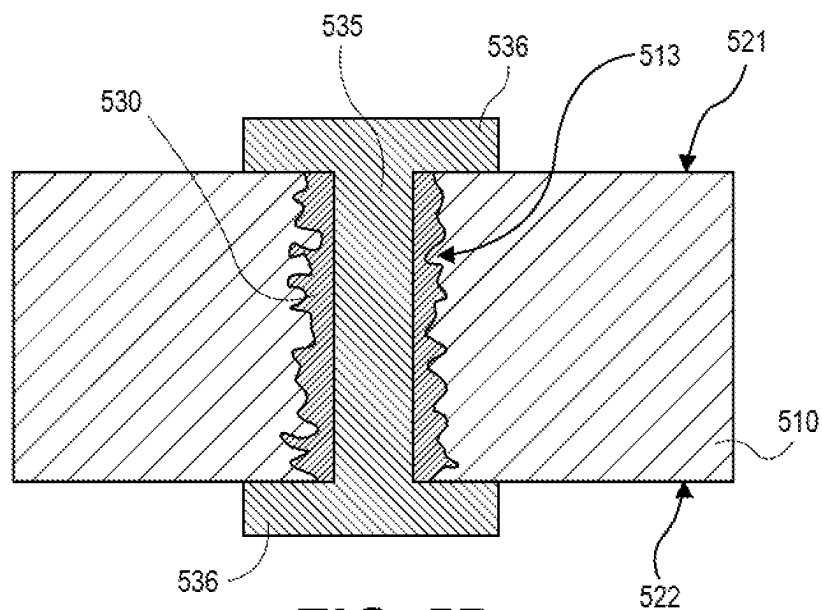


FIG. 5B

**FIG. 5C****FIG. 5D**

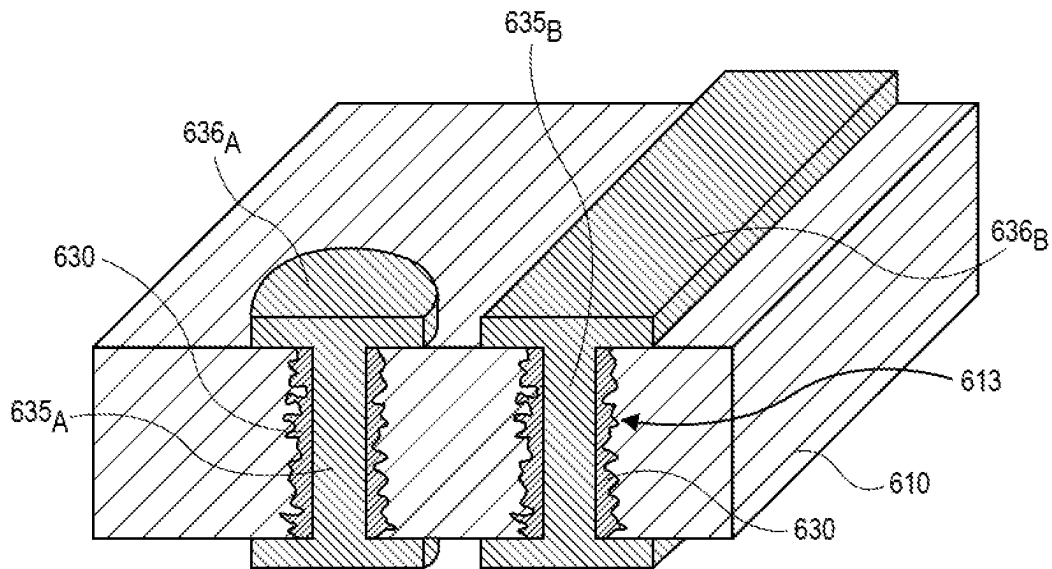


FIG. 6

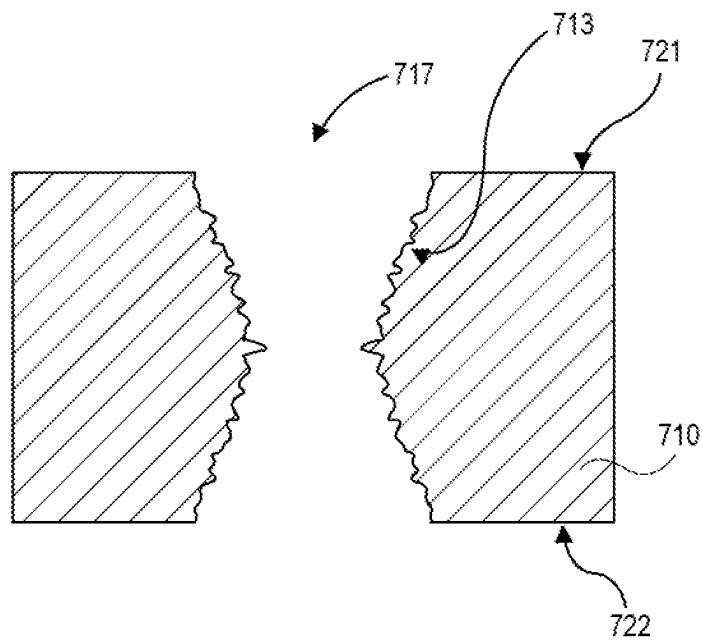


FIG. 7A

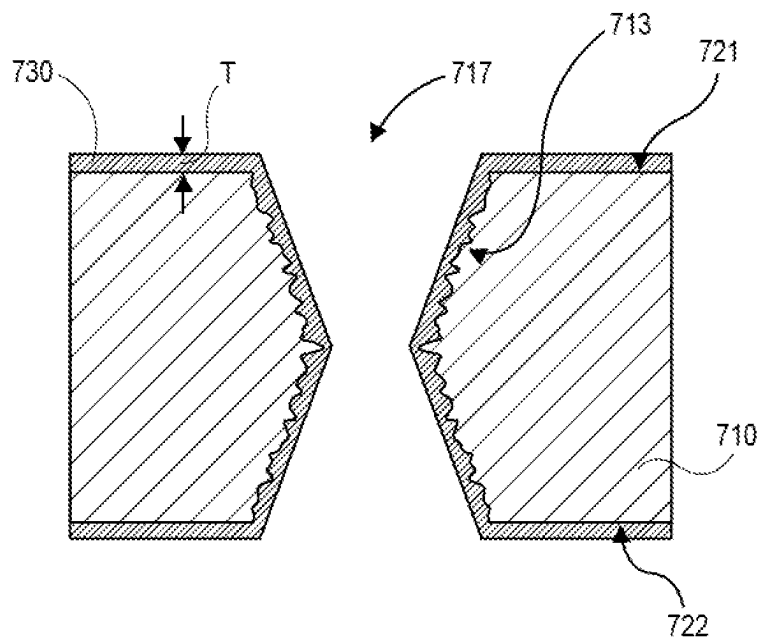


FIG. 7B

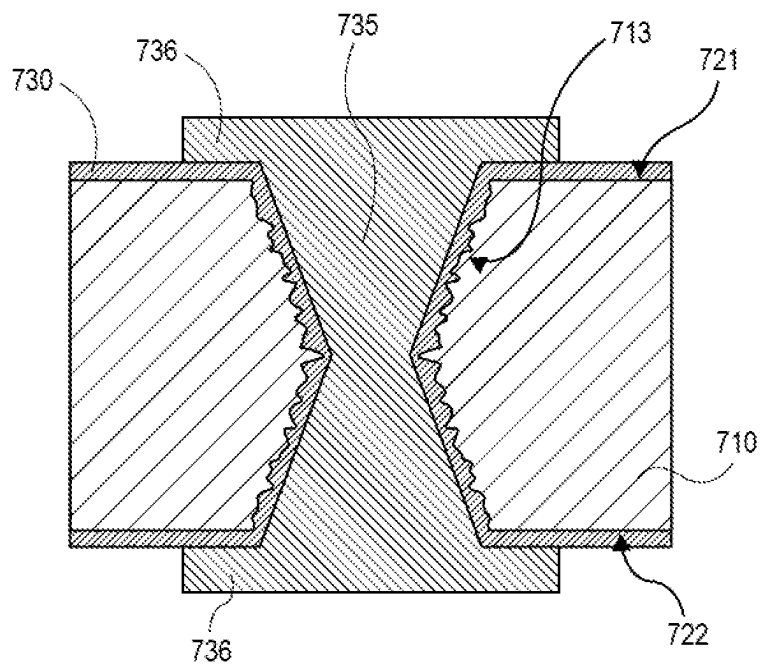


FIG. 7C

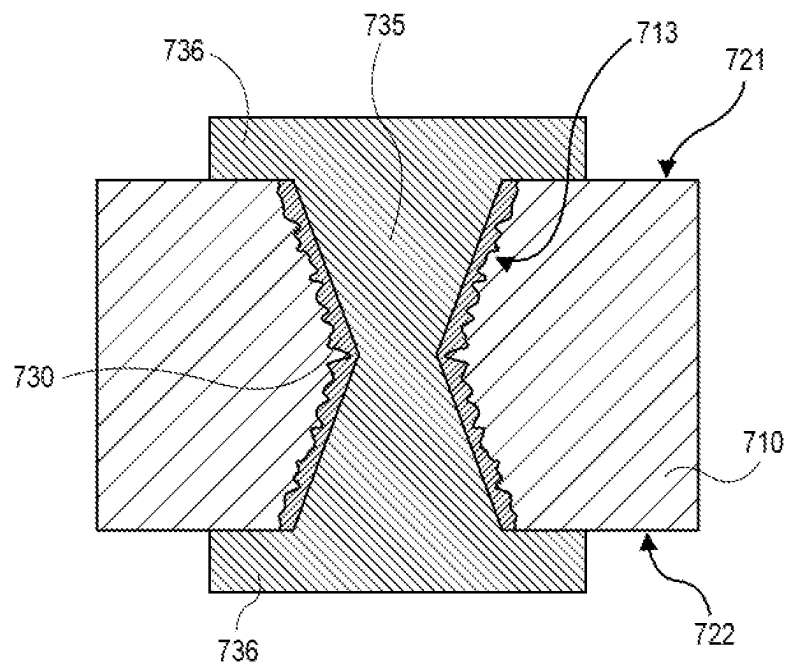


FIG. 7D

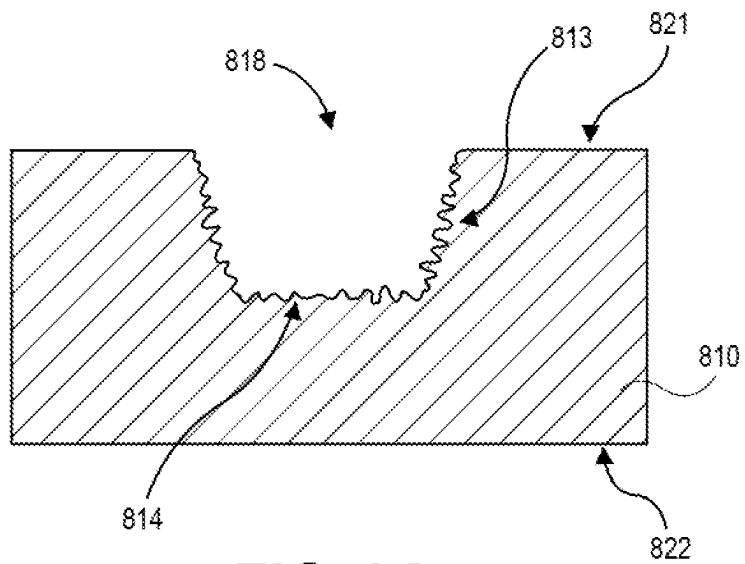


FIG. 8A

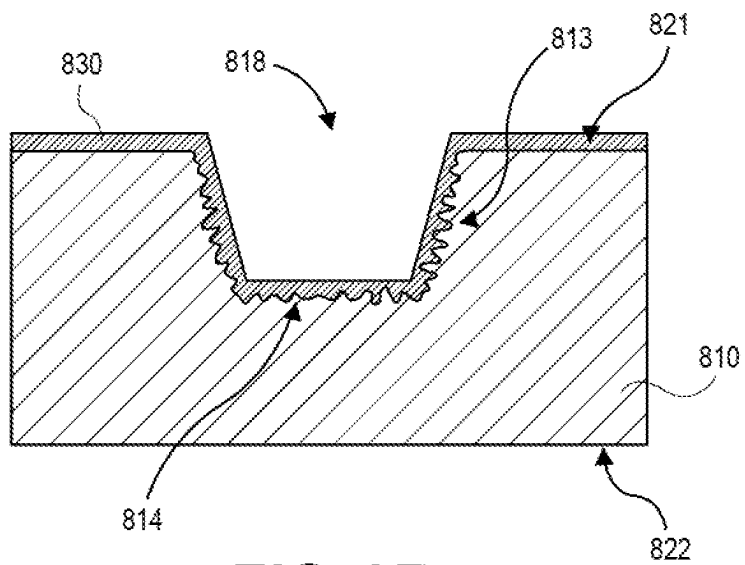


FIG. 8B

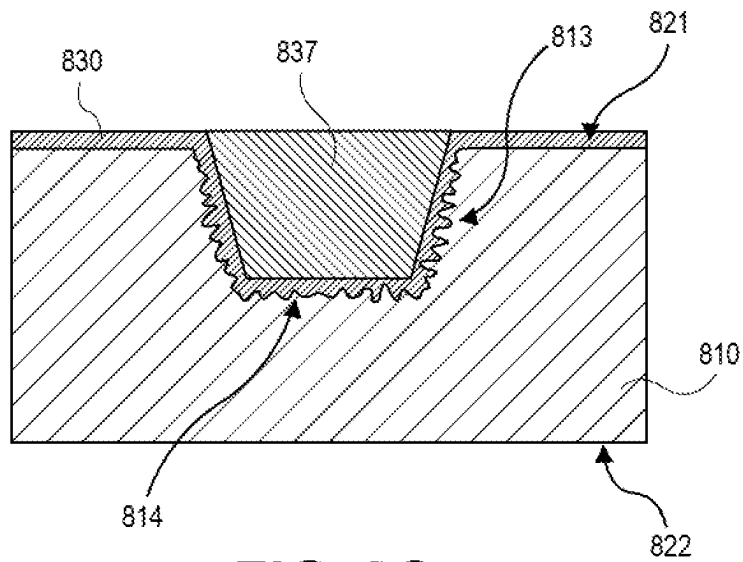


FIG. 8C

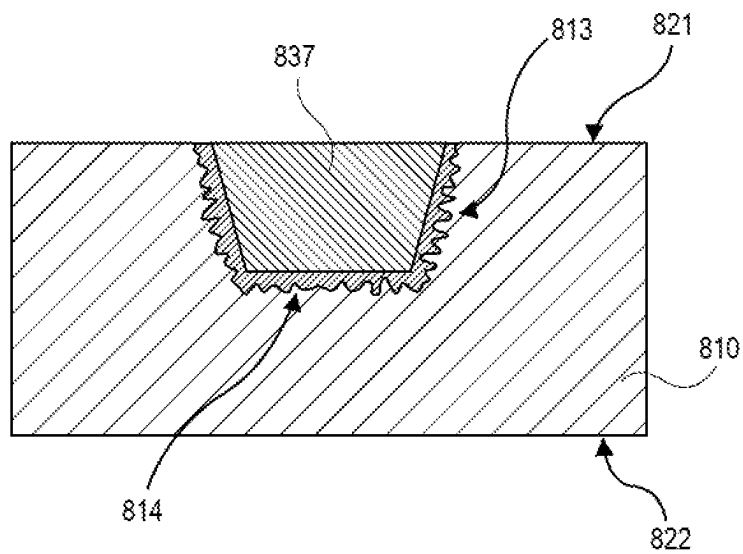
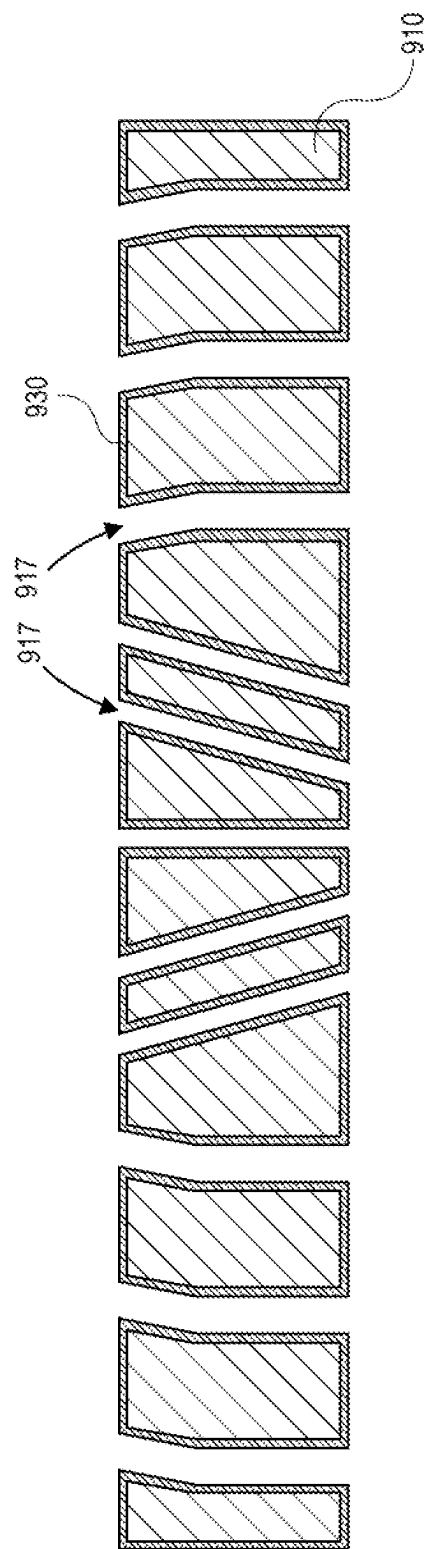
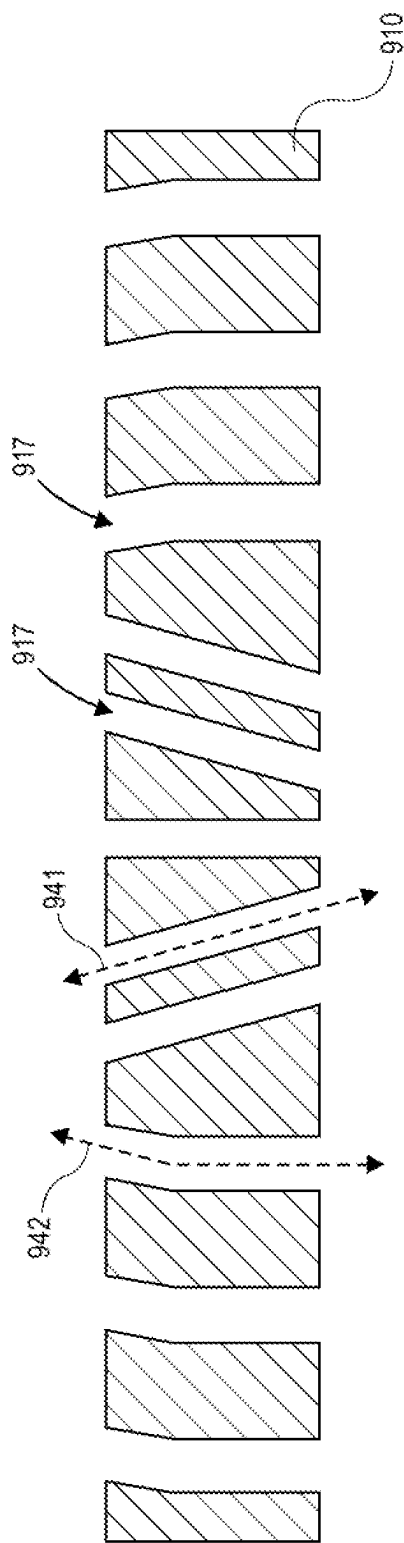


FIG. 8D



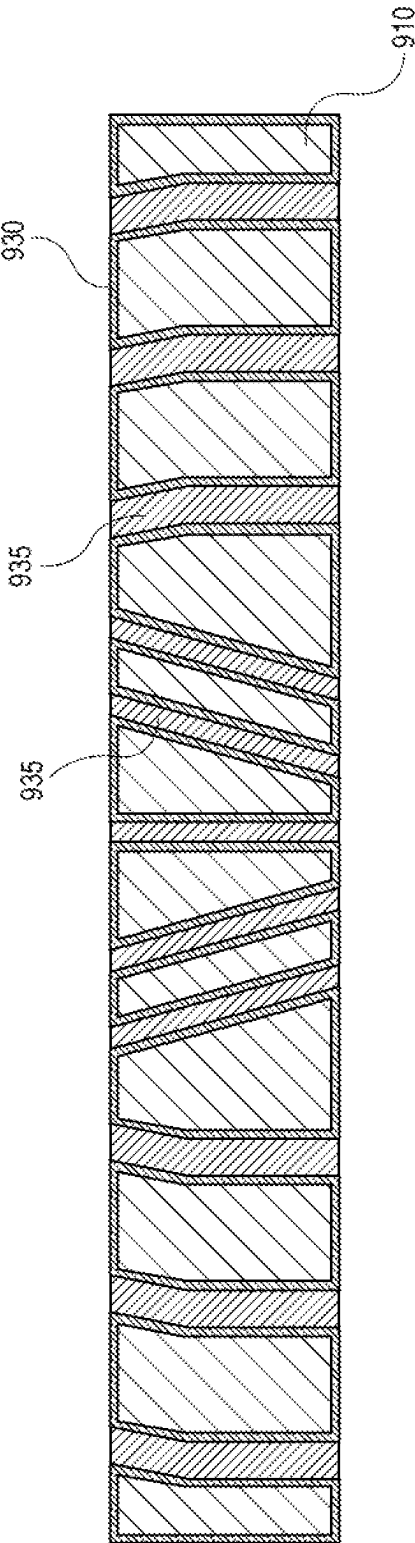
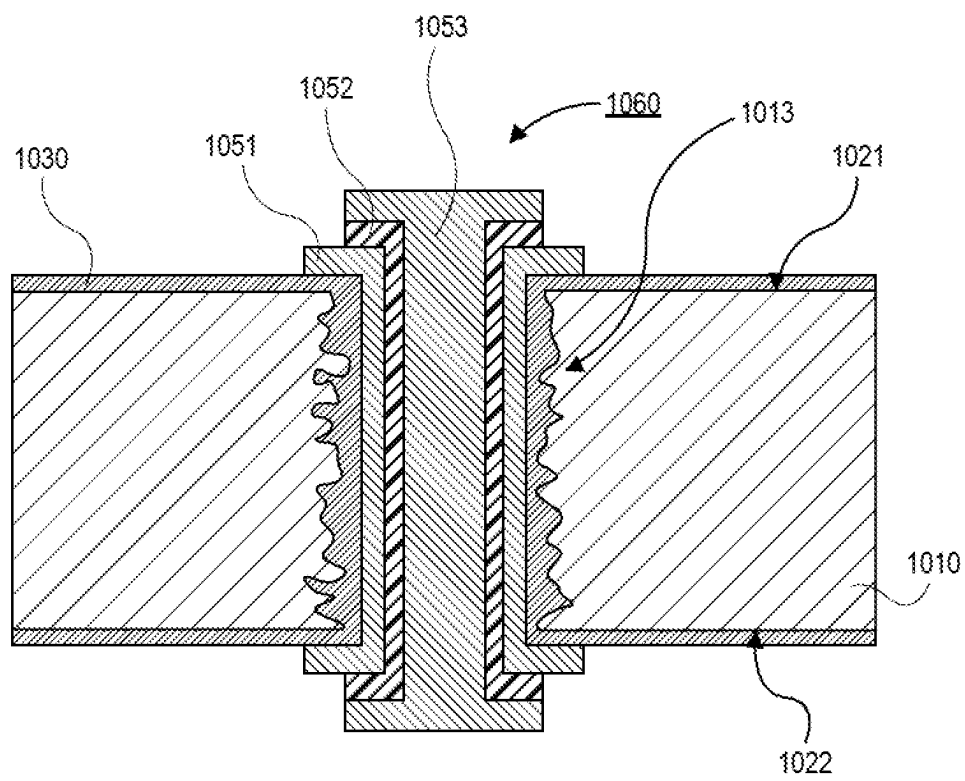


FIG. 9C

**FIG. 10**

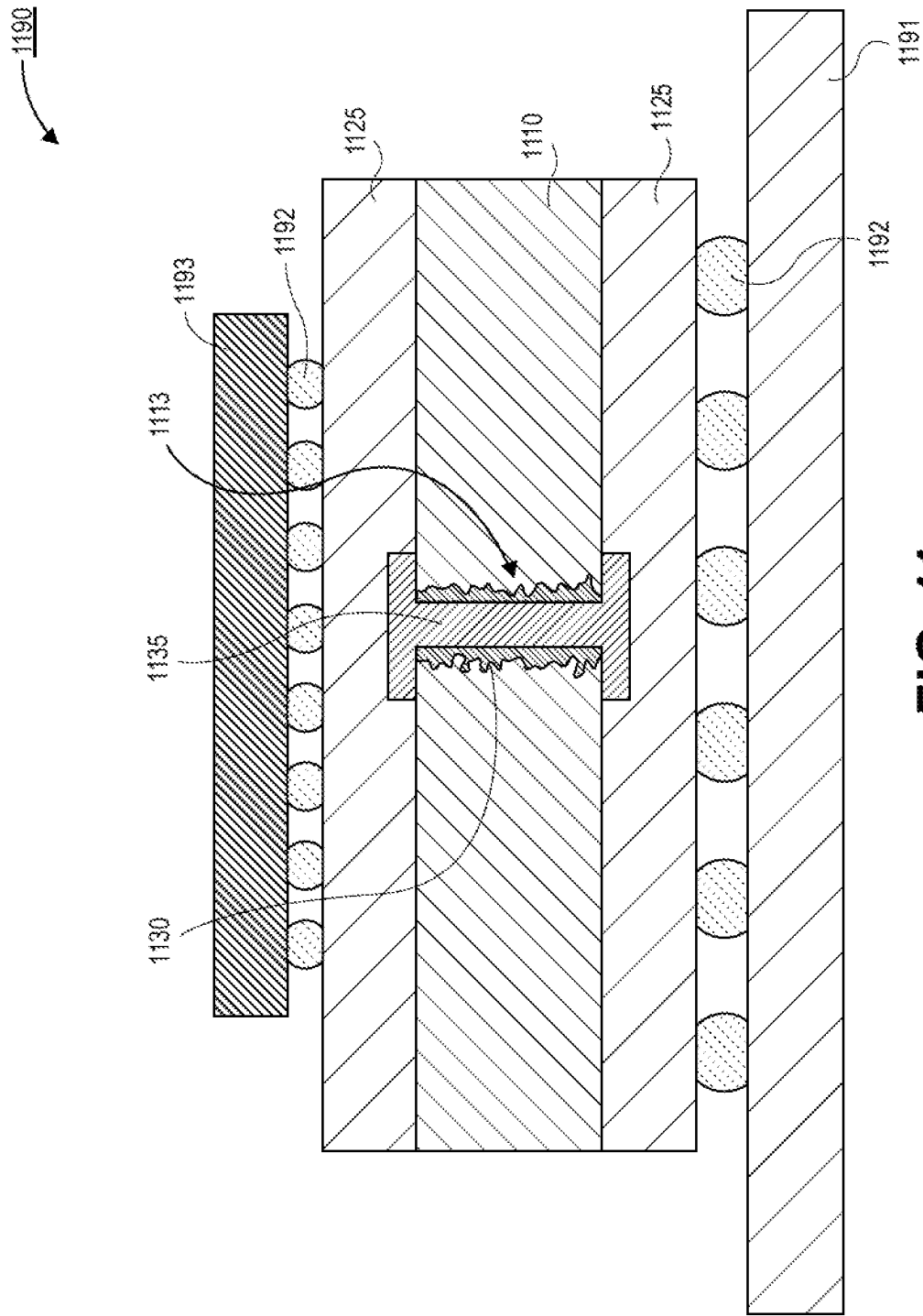
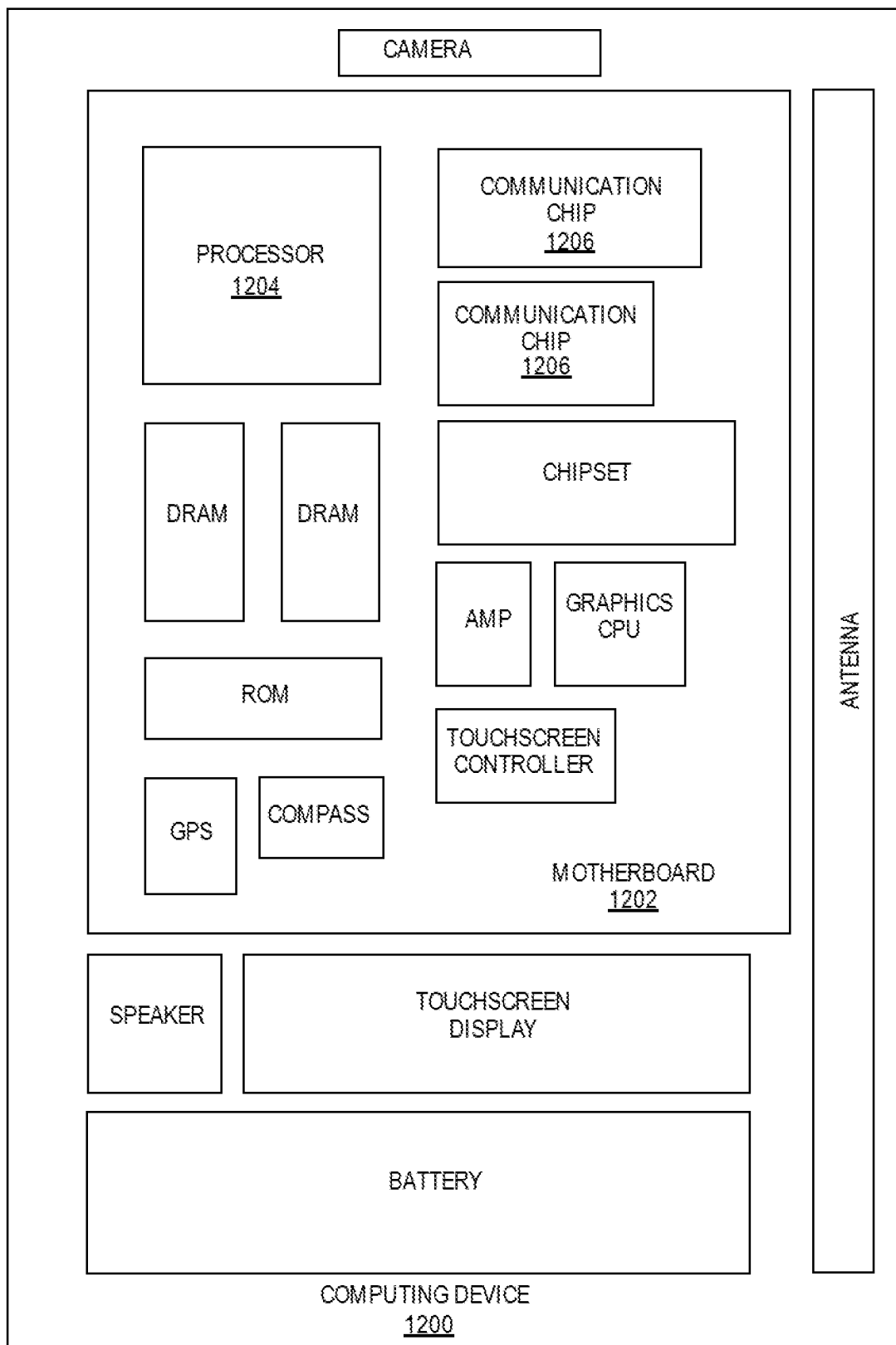


FIG. 11

**FIG. 12**

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DIELECTRIC FILM COATING FOR THROUGH GLASS VIAS AND PLANE SURFACE ROUGHNESS MITIGATION

TECHNICAL FIELD

Embodiments of the present disclosure relate to electronic packages, and more particularly to electronic packages with vias and planes that are lined with a dielectric film to reduce surface roughness.

BACKGROUND

In advanced electronic packaging, some implementations are beginning to use glass cores. The glass core may be etched through with a laser assisted etching process. Generally, a laser assisted etching process involves exposing the glass core with a laser. The laser exposure changes a morphology of the glass material. The change in morphology renders the exposed regions etch selective to the unexposed regions. As such, an etching chemistry (e.g., a wet etching chemistry) can be used to selectively remove the exposed regions of the glass core.

However, the laser assisted etching process may result in rough sidewalls of the via openings in the glass core. For example, root means squared (RMS) roughness of the sidewalls of via openings may be 100 nm or greater, or even 200 nm or greater. Average roughness Ra may also be approximately 100 nm or greater. The roughness of the surfaces reduces electrical performance of the resulting conductive structure provided in the via openings.

BRIEF DESCRIPTION OF THE DRAWINGS

FIGS. 1A-1C are cross-sectional illustrations depicting a laser assisted etching process for forming vias through a core, in accordance with an embodiment.

FIGS. 2A-2C are cross-sectional illustrations depicting a laser assisted etching process for forming blind vias into a core, in accordance with an embodiment.

FIGS. 3A-3C are cross-sectional illustrations depicting a laser assisted etching process for forming a blind via into a core, in accordance with an embodiment.

FIG. 4A is a cross-sectional illustration of a glass core with a via opening, in accordance with an embodiment.

FIG. 4B is a graph of the surface roughness of the via opening in FIG. 4A provided by an atomic force microscope (AFM), in accordance with an embodiment.

FIGS. 5A-5D are cross-sectional illustrations depicting a process for forming a via through a glass core with a liner over the sidewalls of the via opening, in accordance with an embodiment.

FIG. 6 is a perspective view illustration of a glass core with a via and a via plane, where sidewalls through the glass core are lined with a liner, in accordance with an embodiment.

FIGS. 7A-7D are cross-sectional illustrations depicting a process for forming a via through a glass core with via opening sidewalls that are tapered and covered by a liner, in accordance with an embodiment.

FIGS. 8A-8D are cross-sectional illustrations depicting a process for forming a blind via into a glass core with via opening sidewalls that are tapered and covered by a liner, in accordance with an embodiment.

FIGS. 9A-9C are cross-sectional illustrations depicting a process for forming vias that are angled with a liner over the via opening sidewalls, in accordance with an embodiment.

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FIG. 10 is a cross-sectional illustration of a trench capacitor with a liner over the via opening sidewalls, in accordance with an embodiment.

FIG. 11 is a cross-sectional illustration of an electronic system with a package substrate with a glass core and a liner over sidewalls of via openings, in accordance with an embodiment.

FIG. 12 is a schematic of a computing device built in accordance with an embodiment.

EMBODIMENTS OF THE PRESENT DISCLOSURE

Described herein are electronic packages with vias and planes that are lined with a dielectric film to reduce surface roughness, in accordance with various embodiments. In the following description, various aspects of the illustrative implementations will be described using terms commonly employed by those skilled in the art to convey the substance of their work to others skilled in the art. However, it will be apparent to those skilled in the art that the present invention may be practiced with only some of the described aspects. For purposes of explanation, specific numbers, materials and configurations are set forth in order to provide a thorough understanding of the illustrative implementations. However, it will be apparent to one skilled in the art that the present invention may be practiced without the specific details. In other instances, well-known features are omitted or simplified in order not to obscure the illustrative implementations.

Various operations will be described as multiple discrete operations, in turn, in a manner that is most helpful in understanding the present invention, however, the order of description should not be construed to imply that these operations are necessarily order dependent. In particular, these operations need not be performed in the order of presentation.

As noted above, laser assisted etching processes may sometimes be limited by the resulting surface roughness of the via opening sidewalls. For example, root mean squared (RMS) roughness and roughness average (Ra) may both be over 100 nm. In some instances RMS and Ra roughness may be approximately 200 nm or greater. Currently, attempts to reduce surface roughness are limited. For example, glass composition and etching chemistry may be selected to reduce surface roughness. However, even optimal glass compositions and etching chemistries still produce surface roughness values that are not compatible with some through core structures, such as vias, capacitors, and the like.

Accordingly, embodiments disclosed herein include a liner that is disposed over the sidewall surfaces of the via openings. The liner may be disposed with a conformal deposition process. As such, the liner is able to fill the topography produced by the surface roughness. In some embodiments, the liner is deposited with an atomic layer deposition (ALD) process, a chemical vapor deposition (CVD), or physical vapor deposition (PVD) process. The liner fills the topography and produces a new surface that has a lower surface roughness. For example, RMS roughness and/or Ra of the outer surface of the liner may be approximately 50 nm or less. In a particular embodiment, the RMS roughness and/or Ra may be approximately 10 nm or less. As such, subsequently deposited conductor material (e.g., to form a via) may have surfaces with low roughness. The low roughness on the conductive material results in better electrical properties.

In order to provide context, FIGS. 1A-3C are three series of cross-sectional illustrations that depict processes for

forming features in glass cores with laser assisted etching processes. In FIGS. 1A-1C, a through core via opening is formed. In FIGS. 2A-2C a pair of blind via openings on opposite surfaces of the core are formed. In FIGS. 3A-3C a blind via opening into the top surface of the core is formed. The openings formed in FIGS. 1A-3C can then be filled with materials (e.g., liners, conductive materials, etc.) using various plating or other deposition processes in order to manufacture vias and/or planes within a glass core. However, without the presence of a liner, the surface roughness of the sidewalls of the glass core may be too rough for some electrical features in the package.

Referring now to FIGS. 1A-1C, a series of cross-sectional illustrations depicting a process for fabricating openings in a glass core 110 is shown, in accordance with an embodiment.

Referring now to FIG. 1A, a cross-sectional illustration of a glass core 110 is shown, in accordance with an embodiment. In an embodiment, the glass core 110 may have a thickness that is between approximately 50 μm and approximately 1,000 μm . As used herein, approximately may refer to a value that is within 10% of the stated value. For example, approximately 50 μm may refer to a value between 45 μm and 55 μm . Though, it is to be appreciated that other thicknesses (larger or smaller) may also be used for the glass core 110. In an embodiment, a laser 180 is used to expose a region of the glass core 110. As shown in FIG. 1A, the exposure may be made on both sides (i.e., the top surface of the glass core 110 and the bottom surface of the glass core 110). A single laser 180 may be used, or multiple lasers may be used. In an embodiment, the laser 180 is exposed over the glass core 110 at locations where via openings are desired.

Referring now to FIG. 1B, a cross-sectional illustration of the glass core 110 after the laser 180 exposure is completed is shown, in accordance with an embodiment. As shown, the laser 180 exposure may result in the formation of exposed regions 115. In an embodiment, the glass core 110 may comprise a glass material that is able to be morphologically changed upon exposure to a laser 180. For example, the morphological change may result in the microstructure of the glass core 110 transforming to a crystalline structure from an amorphous structure. Accordingly, the exposed region 115 is shown with a different shading than the glass core 110.

In an embodiment, the laser 180 exposure may result in an exposed region 115 that has a tapered sidewall 113. In the instance where both sides of the glass core 110 are exposed (as is the case shown in FIG. 1A), the exposed region 115 may have a double tapered profile. That is, widths of the exposed region 115 at a top surface of the glass core 110 and at a bottom surface of the glass core 110 may be wider than a width at a middle of the glass core 110. In some instances, such a sidewall 113 profile may be referred to as an hourglass shaped profile.

Referring now to FIG. 1C, a cross-sectional illustration of the glass core 110 after the exposed region 115 is removed is shown, in accordance with an embodiment. In an embodiment, removal of the exposed region 115 may result in the formation of a via opening 117. The via opening 117 may pass entirely through a thickness of the glass core 110. In an embodiment, the via opening 117 may be a high aspect ratio via opening 117. As used herein a "high aspect ratio" may refer to an aspect ratio (depth:width) that is approximately 5:1 or greater, with the width being measured at a narrowest point through a thickness of the via opening 117. In other embodiments, the aspect ratio of the via opening 117 may be

approximately 10:1 or greater, approximately 20:1 or greater, or approximately 50:1 or greater.

Referring now to FIGS. 2A-2C, a series of cross-sectional illustrations depicting a process for forming blind structures into a glass core 210 is shown, in accordance with an embodiment. Instead of forming an opening entirely through the glass core 210, structures that extend partially through a thickness of the core 210 are provided.

Referring now to FIG. 2A, a cross-sectional illustration of a glass core 210 is shown, in accordance with an embodiment. In an embodiment, the glass core 210 may be substantially similar to the glass core 110 described in greater detail above. For example, the glass core 210 may have a thickness between approximately 50 μm and approximately 1,000 μm . In an embodiment, lasers 280 may expose portions of the glass core 210. In an embodiment, the laser 280 exposure in FIG. 2A may be different than the laser 180 exposure in FIG. 1A. For example, an intensity or duration of the laser 280 exposure may be less than the intensity or duration of the laser 180 exposure in FIG. 1A.

Referring now to FIG. 2B, a cross-sectional illustration of the glass core 210 after exposed regions 215 are formed is shown, in accordance with an embodiment. In an embodiment, the exposed regions 215 do not extend entirely through a thickness of the glass core 210. For example, a region 218 may be provided between the top exposed region 215 and the bottom exposed region 215. In some instances, the exposed regions 215 still include tapered sidewalls 213. Since the exposed regions 215 are formed from only a single side, the sidewalls 213 may only have a single taper. That is, the exposed regions 215 may not be hourglass shaped.

Referring now to FIG. 2C, a cross-sectional illustration of the glass core 210 after the exposed regions 215 are removed to form openings 217 is shown, in accordance with an embodiment. In an embodiment, the exposed regions 215 may be removed with an etching process that is selective to the exposed regions 215 over the rest of the glass core 210. As shown, the openings 217 do not extend entirely through the glass core 210. In such embodiments, the openings 217 may be referred to as blind openings since they do not pass through the glass core 210.

Referring now to FIGS. 3A-3C, a series of cross-sectional illustrations depicting a process for forming a blind opening 317 is shown, in accordance with an embodiment.

Referring now to FIG. 3A, a cross-sectional illustration of a glass core 310 is shown, in accordance with an embodiment. In an embodiment, the glass core 310 may be substantially similar to the glass cores 110 and 210 described in greater detail above. For example, the glass core 310 may have a thickness between approximately 50 μm and approximately 1,000 μm . In an embodiment, a laser 380 may be used to expose a surface of the glass core 310. In contrast to embodiments described in greater detail above, the laser 380 exposure may only be provided on a single surface of the glass core 310.

Referring now to FIG. 3B, a cross-sectional illustration of the glass core 310 after the laser exposure to form an exposed region 315 is shown, in accordance with an embodiment. In an embodiment, the exposed region 315 may be a region that has a morphology change compared to the rest of the glass core 310. For example, the morphology change may be the transition from an amorphous structure to a crystalline structure. In an embodiment, the exposed region 315 may not extend entirely through a thickness of the glass core 310. That is, the exposed region 315 may be suitable for forming blind structures.

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However, it is to be appreciated that in some embodiments, a laser **380** exposure on a single surface of the glass core **310** can be used to form an exposed region **315** that extends through an entire thickness of the glass core **310**. That is, it is not necessary to use an exposure on both sides of the glass core **310** in order to form through core structures. In such an embodiment, the sidewall profile of the exposed region **315** may have a single taper, instead of the hourglass shaped taper shown in FIG. 1B.

Referring now to FIG. 3C, a cross-sectional illustration of the glass core **310** after the exposed region **315** is removed is shown, in accordance with an embodiment. In an embodiment, the removal of the exposed region **315** may result in an opening **317** being formed into the surface of the glass core **310**. In an embodiment, the opening **317** may be a blind opening. In other embodiments, the opening **317** may pass entirely through a thickness of the glass core **310**.

Referring now to FIG. 4A, a cross-sectional illustration of a glass core **410** is shown, in accordance with an embodiment. In an embodiment, the glass core **410** comprises a first surface **421** and a second surface **422**. A via opening **417** may be provided through a thickness of the glass core **410** from the first surface **421** to the second surface **422**. In an embodiment, the sidewalls **413** of the via opening **417** may be tapered and have an hourglass shaped profile. In the illustration shown in FIG. 4A, the sidewalls **413** are shown as being relatively smooth. However, it is to be appreciated that the etching process used to form the via opening **417** may impart an undesirable surface roughness on the sidewalls **413**.

For example, a plot **420** of the surface roughness on a portion of the sidewall **413** is shown in FIG. 4B. FIG. 4B is an atomic force microscope (AFM) plot of the surface roughness of the sidewall **413**. As shown, the surface roughness can range between approximately 400 nm and 400 nm. Accordingly, despite appearing to be smooth, there may be a significant roughness that needs to be accounted for in order to produce high performance electronic packages.

Referring now to FIGS. 5A-5D, a series of cross-sectional illustrations depicting a process for forming a via in a core is shown, in accordance with an embodiment. In the illustrated embodiments surface roughness of the sidewalls **513** are exaggerated in order to more clearly depict certain aspect of the various embodiments described herein. That is, depending on the degree of magnification used to examine the sidewalls **513**, there may not be any discernable surface roughness.

Referring now to FIG. 5A, a cross-sectional illustration of a core **510** is shown, in accordance with an embodiment. In an embodiment, the core **510** may be a glass core **510**. The core **510** may have a thickness between approximately 50 μm and approximately 1,000 μm . In an embodiment, the core **510** has a first surface **521** (i.e., a top surface) and a second surface **522** (i.e., a bottom surface). In an embodiment, a via opening **517** is formed through a thickness of the core **510** from the first surface **521** to the second surface **522**. In an embodiment, the via opening **517** may be a high aspect ratio feature. For example, the aspect ratio of the via opening **517** may be 10:1 or greater, 20:1 or greater, or 50:1 or greater. At the macro scale, the sidewalls **513** may be substantially vertical. However, at smaller scales, the sidewalls **513** may have a surface roughness, as indicated in FIG. 5A. In an embodiment, the sidewalls **513** may have an RMS roughness and/or a Ra that is approximately 100 nm or greater, or approximately 200 nm or greater.

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Referring now to FIG. 5B, a cross-sectional illustration of the core **510** after deposition of a liner **530** is shown, in accordance with an embodiment. The liner **530** may be deposited over the sidewalls **513**, over the first surface **521**, and over the second surface **522**. In an embodiment, the liner **530** may be deposited with any suitable deposition process, such as ALD, CVD, PVD, spin coated and the like. In a particular embodiment, the liner **530** may be deposited with a conformal deposition process. The liner **530** may have a thickness T that is approximately two to approximately five times the average sidewall **513** surface roughness. In a particular embodiment, the liner **530** may have a thickness T that is approximately 50 nm or greater, or approximately 200 nm or greater. In an embodiment, an outer surface of the liner **530** may have a surface roughness that is less than the surface roughness of the sidewall **513**. In a particular embodiment, the surface roughness of the liner **530** may be approximately 50 nm or less, or approximately 10 nm or less.

The liner **530** may be a dielectric material. For example, the liner **530** may comprise SiO_2 , carbon doped oxides (CDOs), porous CDOs, HfO_x , or Al_2O_3 . In addition to providing improved surface roughness, the liner **530** may also be a material that improves adhesion between the core **510** and a conductive feature disposed in the via opening **517** in a subsequent processing operation. In yet another embodiment, the liner **530** may provide a buffer between the coefficient of thermal expansion (CTE) difference between the core **510** and the conductive feature.

Referring now to FIG. 5C, a cross-sectional illustration of the core **510** after a via **535** is disposed in the via opening **517** is shown, in accordance with an embodiment. In an embodiment, the via **535** interfaces directly with the liner **530**. That is, the via **535** may be spaced away from the surface of the core **510** by the liner **530**. Also, while not shown in FIG. 5C, it is to be appreciated that a seed layer may be provided between the liner **530** and the via **535**. Also, while not shown in FIG. 5C, it is to be appreciated that an adhesion layer may be provided between the liner **530** and the via **535** and/or seed layer (not shown). In some instances the seed layer is the same material as the via **535**. In other embodiments, the seed layer may be a conductor with a composition that is different than the via **535**. Pads **536** may be provided over the top and bottom surface of the via **535**. The pads **536** may be deposited during the same deposition process used to form the via **535** in some embodiments. As shown, the surface of the pads **536** facing the core **510** may be spaced away from the core **510** by a portion of the liner **530**.

However, as shown in FIG. 5D, the liner **530** may be omitted from the first surface **521** and the second surface **522**. For example, a polishing process (e.g., a chemical mechanical polishing (CMP)) may be used to remove the liner **530** from the first surface **521** and the second surface **522**. The polishing process may be implemented before the deposition process used to form the via **535**. In such an embodiment, the surfaces of the pads **536** that face the core **510** may be in direct contact with the core **510**. Although, not shown pads **536** may be in direct contact to either a metal seed layer or adhesion layer (not liner **530**). However, it is to be appreciated that a small portion of the pad **536** may contact the liner **530** at the corner where the pad **536** meets the via **535**.

Referring now to FIG. 6, a perspective view illustration of a core **610** is shown, in accordance with an embodiment. In an embodiment, a first via **635A** and a second via **635B** are shown in the core **610**. The first via **635A** may be a

traditional via that passes through a thickness of the core **610** with a diameter that is smaller than the overlying pad **636A**. The second via **635B** may be considered a via plane. That is, the second via **635B** extends laterally below the length of the overlying pad **636B**. As such, the via **635B** has a plane like structure.

However, despite their structural differences, the first via **635A** and the second via **635B** may share a similar cross-section, as shown at the front face of the core **610**. As shown, both the first via **635A** and the second via **635B** have a liner **630** between the via **635** and the core **610**. The liner **630** may be substantially similar to the liner **530** described in greater detail above. As such, the rough surface of the core **610** is smoothed out to a much lower surface roughness before the vias **635A** and **635B** are deposited into the core **610**.

Referring now to FIGS. 7A-7D, a series of cross-sectional illustrations depicting a process for forming a via with a liner is shown, in accordance with an embodiment. In the illustrated embodiments surface roughness of the sidewalls **713** are exaggerated in order to more clearly depict certain aspect of the various embodiments described herein. That is, depending on the degree of magnification used to examine the sidewalls **713**, there may not be any discernable surface roughness.

Referring now to FIG. 7A, a cross-sectional illustration of a core **710** is shown, in accordance with an embodiment. In an embodiment, the core **710** may be a glass core **710**. The core **710** may have a thickness between approximately 50 μm and approximately 1,000 μm . In an embodiment, the core **710** has a first surface **721** (i.e., a top surface) and a second surface **722** (i.e., a bottom surface). In an embodiment, a via opening **717** is formed through a thickness of the core **710** from the first surface **721** to the second surface **722**. In an embodiment, the via opening **717** may be a high aspect ratio feature. For example, the aspect ratio of the via opening **717** may be 10:1 or greater, 20:1 or greater, or 50:1 or greater. At the macro scale, the sidewalls **713** may have a tapered profile. For example, the macro level taper in FIG. 7A may result in the formation of an hourglass shaped profile for the via opening **717**. However, at smaller scales, the sidewalls **713** may have a surface roughness, as indicated in FIG. 7A. In an embodiment, the sidewalls **713** may have an RMS roughness and/or a Ra that is approximately 100 nm or greater, or approximately 200 nm or greater.

Referring now to FIG. 7B, a cross-sectional illustration of the core **710** after deposition of a liner **730** is shown, in accordance with an embodiment. The liner **730** may be deposited over the sidewalls **713**, over the first surface **721**, and over the second surface **722**. In an embodiment, the liner **730** may be deposited with any suitable deposition process, such as ALD, CVD, PVD, spin coated, slit coated, and the like. In a particular embodiment, the liner **730** may be deposited with a conformal deposition process. The liner **730** may have a thickness T that is approximately two to approximately five times the average sidewall **713** surface roughness. In a particular embodiment, the liner **730** may have a thickness T that is approximately 50 nm or greater, or approximately 200 nm or greater. In an embodiment, an outer surface of the liner **730** may have a surface roughness that is less than the surface roughness of the sidewall **713**. In a particular embodiment, the surface roughness of the liner **730** may be approximately 50 nm or less, or approximately 10 nm or less. Generally, the liner **730** may be a material and/or structure that is similar to the liner **530** described in greater detail above.

Referring now to FIG. 7C, a cross-sectional illustration of the core **710** after a via **735** is disposed in the via opening

717 is shown, in accordance with an embodiment. In an embodiment, the via **735** interfaces directly with the liner **730**. That is, the via **735** may be spaced away from the surface of the core **710** by the liner **730**. Also, while not shown in FIG. 5C, it is to be appreciated that a seed layer may be provided between the liner **730** and the via **735**. In some instances the seed layer is the same material as the via **735**. In other embodiments, the seed layer may be a conductor with a composition that is different than the via **735**. Pads **736** may be provided over the top and bottom surface of the via **735**. The pads **736** may be deposited during the same deposition process used to form the via **735** in some embodiments. As shown, the surface of the pads **736** facing the core **710** may be spaced away from the core **710** by a portion of the liner **730**.

However, as shown in FIG. 7D, the liner **730** may be omitted from the first surface **721** and the second surface **722**. For example, a polishing process (e.g., CMP) may be used to remove the liner **730** from the first surface **721** and the second surface **722**. The polishing process may be implemented before the deposition process used to form the via **735**. In such an embodiment, the surfaces of the pads **736** that face the core **710** may be in direct contact with the core **710**. However, it is to be appreciated that a small portion of the pad **736** may contact the liner **730** at the corner where the pad **736** meets the via **735**. Additionally, in some embodiments, a metal seed layer (not shown) may be provided between the pad **736** and the core **710**.

Referring now to FIGS. 8A-8D, a series of cross-sectional illustrations depicting a process for forming a blind via with a liner is shown, in accordance with an embodiment. In the illustrated embodiments surface roughness of the sidewalls **813** and bottom surface **814** are exaggerated in order to more clearly depict certain aspect of the various embodiments described herein. That is, depending on the degree of magnification used to examine the sidewalls **813** and bottom surface **814**, there may not be any discernable surface roughness.

Referring now to FIG. 8A, a cross-sectional illustration of a core **810** is shown, in accordance with an embodiment. In an embodiment, the core **810** may be substantially similar to the glass cores described in greater detail above. In an embodiment, the core **810** has a first surface **821** (i.e., a top surface) and a second surface **822** (i.e., a bottom surface). In an embodiment, a blind via opening **818** is formed into, but not through, a thickness of the core **810**. At the macro scale, the sidewalls **813** may have a tapered profile. For example, the macro level taper in FIG. 8A may result in the formation of a trapezoidal shaped profile for the blind via opening **818**. However, at smaller scales, the sidewalls **813** and bottom surface **814** may have a surface roughness, as indicated in FIG. 8A. In an embodiment, the sidewalls **813** and bottom surface **814** may have an RMS roughness and/or a Ra that is approximately 100 nm or greater, or approximately 200 nm or greater.

Referring now to FIG. 8B, a cross-sectional illustration of the core **810** after deposition of a liner **830** is shown, in accordance with an embodiment. The liner **830** may be deposited over the sidewalls **813**, the bottom surface **814**, and the first surface **821**. In an embodiment, the liner **830** may be deposited with any suitable deposition process, such as ALD, CVD, PVD, and the like. In a particular embodiment, the liner **830** may have a material and/or structure that is similar to the liners **530** and **730** described in greater detail above.

Referring now to FIG. 8C, a cross-sectional illustration of the core **810** after a blind via **837** is disposed in the via

opening **818** is shown, in accordance with an embodiment. In an embodiment, the blind via **837** interfaces directly with the liner **830**. That is, the blind via **837** may be spaced away from the surface of the core **810** by the liner **830**. Also, while not shown in FIG. **8C**, it is to be appreciated that a seed layer may be provided between the liner **830** and the blind via **837**. The seed layer may be substantially similar to seed layers described in greater detail above.

As shown in FIG. **8D**, the liner **830** may be omitted from the first surface **821**. For example, a polishing process (e.g., CMP) may be used to remove the liner **830** from the first surface **821**. The polishing process may be implemented before the deposition process used to form the blind via **837**.

In the Figures described above, the via openings all had centerlines that were substantially vertical through the thickness of the core. However, embodiments are not limited to such architectures. For example, as shown in FIGS. **9A-9C**, the centerlines of the via openings **917** may be angled or partially angled.

Referring now to FIG. **9A**, a cross-sectional illustration of a core **910** is shown, in accordance with an embodiment. In an embodiment, a plurality of via openings **917** are formed through a thickness of the core **910**. In an embodiment, centerlines **942** and **941** of the via openings **917** may not be entirely vertical. For example, centerlines **942** have an angled top portion that is connected to a vertical bottom portion. Centerlines **941** may be angled through the entire thickness of the core **910**. In FIG. **9A**, the sidewalls of the via openings **917** are shown as being substantially smooth for simplicity. However, it is to be appreciated that the sidewalls of the via openings **917** may have significant surface roughness values, similar to embodiments described in greater detail above.

Referring now to FIG. **9B**, a cross-sectional illustration of the core **910** after a liner **930** is disposed over the exposed surfaces of the core **910** is shown, in accordance with an embodiment. In an embodiment, the liner **930** may be substantially similar to the liners described in greater detail above. Generally, the liner **930** fills the topography of roughened sidewalls of the via openings **917**, and the liner **930** has a smooth outer surface (e.g., 50 nm RMS or less, or 10 nm RMS or less).

Referring now to FIG. **9C**, a cross-sectional illustration of the core **910** after vias **935** are disposed into the via openings **917** is shown, in accordance with an embodiment. As shown, the vias **935** are spaced away from the core **910** by the liner **930**. In some instances, a seed layer (not shown) may also be between the via **935** and the core **910**.

Referring now to FIG. **10**, a cross-sectional illustration of a trench capacitor **1060** is shown, in accordance with an embodiment. In an embodiment, the trench capacitor **1060** is fabricated in a via opening through a core **1010**. As illustrated, sidewalls **1013** of the via opening may have a relatively high roughness. While shown at the macro level in FIG. **10**, it is to be appreciated that the roughness is exaggerated in FIG. **10** in order to illustrate aspects of certain embodiments. The surface roughness of surface **1013** may be substantially similar to surface roughness values of the patterned core surfaces described in greater detail above.

In an embodiment, a liner **1030** is disposed over the first surface **1021** of the core **1010**, the second surface **1022** of the core **1010**, and the sidewall surfaces **1013** of the via opening. The liner **1030** may be substantially similar in structure and composition as liners described in greater detail above. Generally, the liner **1030** conforms to the roughness of the sidewalls **1013** and provides a smooth outer surface onto which the first electrode **1051** is formed. The

first electrode **1051** may comprise copper or the like. In an embodiment, a capacitor dielectric **1052** may be provided over the first electrode **1051**. The capacitor dielectric **1052** may be a high-k dielectric material. As used herein, high-k materials may refer to materials that have a dielectric constant equal to silicon dioxide (SiO₂) or greater. In an embodiment, a second electrode **1053** may be disposed over the capacitor dielectric **1052**. As such, a capacitor **1060** is formed by the first electrode **1051**, the capacitor dielectric **1052**, and the second electrode **1053**. The smooth surface provided by the liner **1030** allows for high break down voltage embedded capacitors. Particularly, the smooth surface enables ultra-thin high-k dielectric layers. The smooth surface reduces the risk for leakage and potential low voltage dielectric breakdown. In the illustrated embodiment, a single via structure is shown. However, it is to be appreciated that the capacitor **1060** may also be formed as a via plane, a blind via, or a plurality of adjacent vias.

Referring now to FIG. **11**, a cross-sectional illustration of an electronic system **1190** is shown, in accordance with an embodiment. In an embodiment, the electronic system **1190** comprises a board **1191**, such as a printed circuit board (PCB). The board **1191** may be coupled to a package substrate by interconnects **1192**. The interconnects **1192** may be solder balls, sockets, or the like. In an embodiment, the package substrate comprises a core **1110** and buildup layers **1125** above and below the core **1110**. The core **1110** may be a glass core, similar to the glass cores described above. In an embodiment, a liner **1130** may line a rough sidewall **1113** of the core **1110**. A via **1135** may be provided over the liner **1130**. The liner **1130** may be substantially similar to any of the liners described in greater detail above. In an embodiment, a die **1193** is coupled to the top buildup layer **1125** by interconnects **1192**. The interconnects **1192** may be any suitable first level interconnect (FLI). In an embodiment, the die **1193** may be a processor, a graphics processor, a memory die, or any other computational die.

FIG. **12** illustrates a computing device **1200** in accordance with one implementation of the invention. The computing device **1200** houses a board **1202**. The board **1202** may include a number of components, including but not limited to a processor **1204** and at least one communication chip **1206**. The processor **1204** is physically and electrically coupled to the board **1202**. In some implementations the at least one communication chip **1206** is also physically and electrically coupled to the board **1202**. In further implementations, the communication chip **1206** is part of the processor **1204**.

These other components include, but are not limited to, volatile memory (e.g., DRAM), non-volatile memory (e.g., ROM), flash memory, a graphics processor, a digital signal processor, a crypto processor, a chipset, an antenna, a display, a touchscreen display, a touchscreen controller, a battery, an audio codec, a video codec, a power amplifier, a global positioning system (GPS) device, a compass, an accelerometer, a gyroscope, a speaker, a camera, and a mass storage device (such as hard disk drive, compact disk (CD), digital versatile disk (DVD), and so forth).

The communication chip **1206** enables wireless communications for the transfer of data to and from the computing device **1200**. The term "wireless" and its derivatives may be used to describe circuits, devices, systems, methods, techniques, communications channels, etc., that may communicate data through the use of modulated electromagnetic radiation through a non-solid medium. The term does not imply that the associated devices do not contain any wires, although in some embodiments they might not. The com-

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munication chip **1206** may implement any of a number of wireless standards or protocols, including but not limited to Wi-Fi (IEEE 802.11 family), WiMAX (IEEE 802.16 family), IEEE 802.20, long term evolution (LTE), Ev-DO, HSPA+, HSDPA+, HSUPA+, EDGE, GSM, GPRS, CDMA, TDMA, DECT, Bluetooth, derivatives thereof, as well as any other wireless protocols that are designated as 3G, 4G, 5G, and beyond. The computing device **1200** may include a plurality of communication chips **1206**. For instance, a first communication chip **1206** may be dedicated to shorter range wireless communications such as Wi-Fi and Bluetooth and a second communication chip **1206** may be dedicated to longer range wireless communications such as GPS, EDGE, GPRS, CDMA, WiMAX, LTE, Ev-DO, and others.

The processor **1204** of the computing device **1200** includes an integrated circuit die packaged within the processor **1204**. In some implementations of the invention, the integrated circuit die of the processor may be part of an electronic package that comprises a glass core with a via opening that is lined with a liner and filled with a via, in accordance with embodiments described herein. The term "processor" may refer to any device or portion of a device that processes electronic data from registers and/or memory to transform that electronic data into other electronic data that may be stored in registers and/or memory.

The communication chip **1206** also includes an integrated circuit die packaged within the communication chip **1206**. In accordance with another implementation of the invention, the integrated circuit die of the communication chip may be part of an electronic package that comprises a glass core with a via opening that is lined with a liner and filled with a via, in accordance with embodiments described herein.

The above description of illustrated implementations of the invention, including what is described in the Abstract, is not intended to be exhaustive or to limit the invention to the precise forms disclosed. While specific implementations of, and examples for, the invention are described herein for illustrative purposes, various equivalent modifications are possible within the scope of the invention, as those skilled in the relevant art will recognize.

These modifications may be made to the invention in light of the above detailed description. The terms used in the following claims should not be construed to limit the invention to the specific implementations disclosed in the specification and the claims. Rather, the scope of the invention is to be determined entirely by the following claims, which are to be construed in accordance with established doctrines of claim interpretation.

Example 1: an electronic package, comprising: a substrate with a first surface and a second surface, wherein the substrate comprises glass; a via opening through the substrate, wherein sidewalls of the via opening have a root mean squared (RMS) surface roughness that is approximately 100 nm or greater; a liner over the sidewalls of the via opening, wherein an RMS surface roughness of the liner is approximately 50 nm or smaller; and a via through the via opening.

Example 2: the electronic package of Example 1, wherein the RMS surface roughness of the liner is approximately 10 nm or smaller.

Example 3: the electronic package of Example 1 or Example 2, wherein a thickness of the liner is up to approximately five times the RMS surface roughness of the sidewalls of the via opening.

Example 4: the electronic package of Example 3, wherein the thickness of the liner is approximately 200 nm or greater.

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Example 5: the electronic package of Examples 1-4, wherein the sidewalls of the via opening are substantially vertical.

Example 6: the electronic package of Examples 1-4, wherein the sidewalls of the via opening are tapered.

Example 7: the electronic package of Example 6, wherein the sidewalls of the via opening have an hourglass shaped profile.

Example 8: the electronic package of Examples 1-7, wherein the liner is a dielectric material.

Example 9: the electronic package of Example 8, wherein the liner comprises SiO₂, carbon doped oxides (CDO), porous CDOs, HfO_x, Al₂O₃, or polymer.

Example 10: the electronic package of Examples 1-9, further comprising: a seed layer between the via and the liner.

Example 11: the electronic package of Examples 1-10, wherein the liner is further provided over the first surface and the second surface of the substrate.

Example 12: the electronic package of Examples 1-11, wherein the via opening passes entirely through a thickness of the substrate from the first surface to the second surface.

Example 13: the electronic package of Examples 1-12, wherein the via opening is a blind feature that does not pass entirely through a thickness of the substrate.

Example 14: the electronic package of Examples 1-13, wherein the via is a via plane.

Example 15: an electronic package, comprising: a substrate with a first surface and a second surface, wherein the substrate comprises glass; a via opening through a thickness of the substrate from the first surface to the second surface; a liner over sidewalls of the via opening; a first conductor over the liner; a dielectric layer over the first conductor; and a second conductor over the dielectric layer.

Example 16: the electronic package of Example 15, wherein a root mean squared (RMS) surface roughness of the sidewalls of the via opening is approximately 100 nm or greater.

Example 17: the electronic package of Example 16, wherein a thickness of the liner is approximately 100 nm or greater.

Example 18: the electronic package of Examples 15-17, wherein an RMS surface roughness of the liner is approximately 10 nm or less.

Example 19: the electronic package of Examples 15-18, wherein the first conductor, the dielectric layer, and the second conductor form a capacitor.

Example 20: the electronic package of Examples 15-19, wherein the sidewalls of the via opening are substantially vertical.

Example 21: the electronic package of Examples 15-20, wherein a thickness of the dielectric layer is approximately 100 nm or less.

Example 22: the electronic package of Examples 15-21, wherein the second conductor has an I-shaped cross-section.

Example 23: an electronic system, comprising: a board; a package substrate coupled to the board, wherein the package substrate, comprises: a substrate with a first surface and a second surface, wherein the substrate comprises glass; a via opening through the substrate, wherein sidewalls of the via opening have a root mean squared (RMS) surface roughness that is approximately 100 nm or greater; a liner over the sidewalls of the via opening, wherein an RMS surface roughness of the liner is approximately 50 nm or smaller; and a via through the via opening; and a die coupled to the package substrate.

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Example 24: the electronic system of Example 23, wherein the RMS surface roughness of the liner is approximately 10 nm or smaller.

Example 25: the electronic system of Example 23 or Example 24, wherein the liner comprises SiO₂, carbon doped oxides (CDO), porous CDOs, HfO_x, Al₂O₃, or polymer.

What is claimed is:

1. An electronic package, comprising:
 - a substrate with a first surface and a second surface, wherein the substrate comprises glass;
 - a via opening through the substrate, wherein sidewalls of the via opening have a root mean squared (RMS) surface roughness that is approximately 100 nm or greater;
 - a liner over the sidewalls of the via opening, wherein an RMS surface roughness of the liner is approximately 50 nm or smaller; and
 - a via through the via opening.
2. The electronic package of claim 1, wherein the RMS surface roughness of the liner is approximately 10 nm or smaller.
3. The electronic package of claim 1, wherein a thickness of the liner is up to approximately five times the RMS surface roughness of the sidewalls of the via opening.
4. The electronic package of claim 3, wherein the thickness of the liner is approximately 200 nm or greater.
5. The electronic package of claim 1, wherein the sidewalls of the via opening are substantially vertical.
6. The electronic package of claim 1, wherein the sidewalls of the via opening are tapered.
7. The electronic package of claim 6, wherein the sidewalls of the via opening have an hourglass shaped profile.
8. The electronic package of claim 1, wherein the liner is a dielectric material.
9. The electronic package of claim 8, wherein the liner comprises SiO₂, carbon doped oxides (CDO), porous CDOs, HfO_x, Al₂O₃, or polymer.

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10. The electronic package of claim 1, further comprising: a seed layer between the via and the liner.

11. The electronic package of claim 1, wherein the liner is further provided over the first surface and the second surface of the substrate.

12. The electronic package of claim 1, wherein the via opening passes entirely through a thickness of the substrate from the first surface to the second surface.

13. The electronic package of claim 1, wherein the via opening is a blind feature that does not pass entirely through a thickness of the substrate.

14. The electronic package of claim 1, wherein the via is a via plane.

15. An electronic system, comprising:

- a board;
- a package substrate coupled to the board, wherein the package substrate, comprises:
 - a substrate with a first surface and a second surface, wherein the substrate comprises glass;
 - a via opening through the substrate, wherein sidewalls of the via opening have a root mean squared (RMS) surface roughness that is approximately 100 nm or greater;
 - a liner over the sidewalls of the via opening, wherein an RMS surface roughness of the liner is approximately 50 nm or smaller; and
 - a via through the via opening; and
- a die coupled to the package substrate.

16. The electronic system of claim 15, wherein the RMS surface roughness of the liner is approximately 10 nm or smaller.

17. The electronic system of claim 15, wherein the liner comprises SiO₂, carbon doped oxides (CDO), porous CDOs, HfO_x, Al₂O₃, or polymer.

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